

TABLE OF CONTENTS

E-410 HIGH VOLTAGE CIRCUIT BREAKERS

	Page No.
1. SCOPE OF WORK.....	E-410-2
2. CODES AND STANDARDS	E-410-2
2.1 Indonesian National Standards.....	E-410-2
2.2 International Electrotechnical Commission	E-410-2
2.3 National Electrical Manufacturers Association.....	E-410-2
3. OPERATING AND DESIGN CONDITIONS	E-410-2
4. DETAILED REQUIREMENTS	E-410-2
4.1 General.....	E-410-2
4.2 Ratings	E-410-3
4.3 Insulating Medium	E-410-5
4.4 Operating Mechanism.....	E-410-6
4.5 Auxiliary Switches.....	E-410-7
4.6 Control Housing.....	E-410-7
4.7 Control Wiring	E-410-8
4.8 Current Transformers.....	E-410-8
4.9 Accessories and Fittings	E-410-8
4.10 Corona.....	E-410-9
4.11 Porcelain	E-410-9
4.12 Nameplates	E-410-9
5. PERFORMANCE TESTING AND GUARANTEES.....	E-410-9
6. QUALITY ASSURANCE/QUALITY CONTROL	E-410-10

E-410 HIGH VOLTAGE CIRCUIT BREAKERS

1. SCOPE OF WORK

The work shall include the design, manufacturing, supply, factory testing, delivery to site, installation, commissioning and testing of 500 kV circuit breakers in accordance with Part 4, Subsections 1.2.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Section 2.5 Codes and Standards, and the following specific standards:

2.1 Indonesian National Standards

- PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)
- SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60060 : High-voltage test techniques
- IEC 60137 : Insulated bushing for alternating voltages above 1000V
- IEC 60617 : Graphical symbols for diagrams
- IEC 62155 : Hollow pressurized and unpressurized ceramic and glass insulators for use in electrical equipment with rated voltages greater than 1000 V
- IEC 62271 : High-voltage switchgear and controlgear

2.3 National Electrical Manufacturers Association

- NEMA 107 : Methods of Measurements of Radio Influence Voltage (RIV) of High Voltage Apparatus

3. OPERATING AND DESIGN CONDITIONS

Circuit breakers shall be designed for outdoor operation as a part of the conventional substation application located in a tropical climate along the seacoast.

4. DETAILED REQUIREMENTS

4.1 General

The main high voltage current carrying parts shall be able to carry the rated current continuously without exceeding the temperatures specified in IEC 62271.

The completely assembled circuit breaker shall be so designed that when the open circuit breaker is subject to impulse, 50 Hz, or switching surge voltage the electric breakdown shall occur to ground and not across the open circuit breaker gap nor to adjacent phases.

Circuit breakers shall be capable of interrupting their circuits for the duty cycle specified. Breakers shall be substantially in the same mechanical and electrical condition at the termination as at the beginning of the specified duty cycle, shall not have emitted flame, oil, or given other indications of failure and shall be capable of carrying their rated voltage and current.

Circuit breakers shall be capable of interrupting their rated interrupting current after trip coil has been energized within the specified time.

The contacts shall be self-wiping and self-cleaning type.
Closing resistor shall be provided as necessary.

4.2 Ratings

The circuit breakers, complete with insulators and accessories, shall have the following ratings.

- (1) Nominal voltage.....500 kV
- (2) Rated maximum voltage525 kV
- (3) Frequency50 Hz
- (4) Number of poles3
- (5) Current ratings:
 - a. Continuous current.....**4,000 A**
 - b. Short circuit breaking current**50 kA**
 - c. Momentary current**125 kA rms**
 - d. Short time (1 second) current.....**50 kA**
- (6) Phase-to-ground insulation:

ADD1
-IIE
-003

- a. Impulse withstand voltage $0.2 \times 50 \mu\text{s}$ crest 1,800 kV
 - b. One minute, 50 Hz , dry (and wet) withstand test voltage 780 kV
 - c. Switching impulse withstand test voltage 1,175 kV
- (7) Open gap insulation:
- a. Impulse withstand voltage ($1.2 \times 50 \mu\text{s}$) crest1,800 (+ 300) kV
 - b. One minute, 50 Hz , dry (and wet) withstand test voltage 760 kV
 - c. Switching impulse withstand test voltage900 (+ 430) kV
- (8) Radio Influence Voltage (RIV)
- The maximum RIV at 1 MHz on circuit breaker (with or without resistors and open or closed) shall be:
- a. Applied voltage phase-to-ground335 kV
 - b. R.I.V.2,500 μ volts
- Measurement shall be made in accordance with NEMA107. In making the RIV test, radio noise readings shall be taken at various points so as to determine the break point.
- (9) Rated interrupting time 50 ms
 - (10) Rated permissible tripping delay 1 sec
 - (11) Reclosing time (adjustable).....0.3 to 2.0 sec
 - (12) Operating duty cycle.....0-0.3 sec-CO-180 sec-CO
 - (13) First pole to clear factor 1.3

The 500 kV breakers shall meet the following Transient Recovery Voltage requirement for the rated short circuit conditions (values specified are referred to representation by four (4) parameters method in accordance with IEC 62271, however, the voltage magnitudes are higher than the IEC recommendations).

$$\begin{aligned}
 U_1 &= 557 \text{ kV} \\
 t_1 &= 279 \mu\text{s (microsecond)} \\
 U_c &= 968 \text{ kV} \\
 t_2 &= 837 \mu\text{s (microsecond)}
 \end{aligned}$$

$t_d = 2 \mu s$ (microsecond)

$U' = 279 \text{ kV}$

$t' = 141 \mu s$ (microsecond)

Surge suppression by reinsertion resistor: (if required)

- (1) The breakers shall use resistor reinsertion to control closing surges for all breakers associated with transmission line switching. Resistor shall be sized according to the following criteria:

- a. Resistor Ohmic value800 Ω *)
- b. Reinsertion duration10 ms (milliseconds)

Note: *) The Contractor shall confirm this parameter at detailed design stage based on Transient Network Analyzer (TNA) study performed by the Contractor.

- (2) The resistor's thermal capability shall be base upon maximum short circuit conditions at the breaker for the duty cycle specified in this Specification.

The 500 kV breaker shall be capable of synchronizing switching to maximum rated MVA generator transformer.

Out of phase switching requirement:

The reference restricting voltage test waveform shall be for a 25 % breaker rating.

The circuit breakers shall be rated for the following line-charging current interrupting capability in accordance with IEC 62271:

500 kV500 A

The 500 kV breaker shall have a normal frequency recovery voltage withstand capability of 3.0 per unit across the open contacts in accordance with relevant standards.

4.3 Insulating Medium

Circuit breakers shall be of the SF₆ insulating gas, single pressure type.

Provisions shall be provided for pulling a vacuum of 8 millibars absolute or less, in less than 4 hours and hold for required drying period prior to filling with insulating medium.

4.4 Operating Mechanism

4.4.1 General

The operating mechanism shall be of the hydraulic, spring or pneumatic type.

The nominal control voltage for closing and tripping shall be 220 V DC. The ranges of the control voltages shall be 187 to 242 V DC for closing and shall be 154 to 242 V DC for tripping. The Contractor shall provide 220 V AC UPS to supply new 500 kV circuit breakers if existing UPS is not sufficient.

If an air compressor, gas compressor or hydraulic pump is furnished, the power supply shall be 400/230 V AC, 3 phase, 4 wire, 50 Hz with suitable over current and short circuit protection. Each compressor shall be furnished with running time meter.

4.4.2 Circuit Breaker Operation

For 500 kV circuit breakers, operating mechanism shall be provided for single pole switching. Each operating mechanism shall have two (2) electrically independent trip coils and dual pressure interlock switches. The two (2) trip coils including the associated circuit and interlocks shall be wired independent and completely redundant.

The pole disagreement relay shall be provided for each independent tripping system and shall have an adjustable pick-up delay of 0.02 - 2 seconds. The contacts shall trip all coils of both independent trip systems.

Space heaters with necessary control devices shall be furnished and mounted in each mechanism housing.

Alarm contacts shall be provided wherever failing of any heaters may result in faulty operation of the breaker or its components.

All devices and contacts shall be available for convenient inspection and maintenance.

nance.

A manually-operated emergency tripping device shall be provided as part of the mechanisms and it shall be protected against accidental operation. The emergency-tripping device shall be capable of tripping the circuit breaker during a complete loss of the AC and DC auxiliary power.

Circuit breaker mechanisms shall be provided with indicators or targets for each operating mechanism. They shall be located so the position of the poles, whether open or closed, can be easily determined.

The operating mechanism and its control relays shall be mechanical and electrical trip-free with anti pump features, and shall be so connected that the closing mechanism will automatically become inoperative after and while the breakers is closed, regardless of the position of the control switch, but will not become inoperative until after the breakers has had ample time to complete the operation of closing. It shall be impossible to operate the closing mechanism again until the operator has first moved the breaker control to the neutral position.

Operating mechanisms shall be provided with alarm contacts to close on low pressure and a low-pressure cut-out to prevent operation of the breakers closing valves with insufficient pressure.

Controls relays, where practical, shall be arranged to break both polarities of the control circuit.

4.5 Auxiliary Switches

Auxiliary switches shall be supplied with each operating mechanism. In addition to the above auxiliary switches, each circuit breaker shall be provided with special auxiliary switches for automatic reclosing service. These auxiliary switches shall have a provision for adjustment in the reclosing time, or in lieu thereof a time delay relay (adjustable) to perform the function.

4.6 Control Housing

Suitable weather-tight cabinets (IPW 55) of liberal size shall be provided for housing the operating mechanisms, relays and terminal blocks.

4.7 Control Wiring

The control, indicating and interlocking wires for the operating mechanism shall be enclosed and shall be brought together to screw type terminal blocks in the operating mechanism cabinets and arranged for iron conduit connection. Secondary wiring from bushing current transformers, which terminates within mechanism cabinet, shall be completely enclosed in iron conduits with watertight joints, between the transformer housing and the main mechanism housing.

4.8 Current Transformers

Current transformers with multiratio bushing type shall be provided as required for the proper metering and protection.

All secondary tap leads of the current transformer shall be brought together in the main breaker mechanism housing to short circuiting type terminal blocks for connection to external cables.

Bushings or terminals shall be provided with conduit entrance for current transformer secondary. All secondary leads shall be wired through steel conduit shorting type barrier terminal blocks in the mechanism housing.

4.9 Accessories and Fittings

Circuit breakers shall be equipped with a complete set of necessary accessories for operation and maintenance. The following shall be included with one (1) per pole where operating mechanisms are on a “per pole” basis:

- (1) Position indicator.
- (2) Operation counter.
- (3) Means for effectively grounding tanks, mechanism housing, etc.
- (4) Maintenance closing device.
- (5) Pressure gauges.
- (6) Pressure switches.
- (7) Low voltage relays for monitoring the power supply to each compressor motor, heaters and closing and tripping circuits. Two (2) C-form contacts

rated 220 volts DC are required from each low voltage relay.

- (8) A complete charge of SF₆ gas for any gas-filled portion of the circuit breaker.
- (9) Pressure switches to detect loss of SF₆ gas pressure or air pressure in any one (1) of the three (3) poles.

4.10 Corona

No visual corona shall exist at the applied voltage. Also the lights shall be turned off in the test bay and visual corona start and stop voltages shall be recorded with the corresponding radio noise voltages on both ascending and descending voltage values. Photographs shall be taken for each recorder voltage values. NEMA 107 test procedures shall be used.

4.11 Porcelain

The Contractor shall meet the requirements of IEC 62155 for all porcelain used in these circuit breakers. Porcelain insulation shall be chocolate brown color.

4.12 Nameplates

Nameplates shall be furnished in accordance with IEC 62271.

5. PERFORMANCE TESTING AND GUARANTEES

Normal type test and special type test for IEC 62271 shall have been performed on a sample of each type of circuit breakers of identical design and the test results shall be submitted.

Routine tests as per IEC 62271 shall be carried out as outlined below.

- (1) Power frequency voltage dry tests of the main circuit of each voltage level shall be performed on 10% of each design units with a minimum of two (2) units tested.
- (2) Voltage tests on control and auxiliary circuits shall be performed on all units.
- (3) Measurement of the resistance of the main circuit samples are same as clause (1) "power frequency voltage dry tests" above of this Specification.
- (4) Mechanical operating tests shall be performed on all units.

- (5) Mechanical/hydraulic pressure tests on all containers under pressure to twice the maximum operating pressure for static pressures and three (3) times the maximum operating pressure for containers subject to impulse or rapid pressure changes.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control of the high-voltage circuit breakers shall conform to the requirements included in Part 4, Section 2, Subsection 2.7.

The quality assurance/quality control procedures as specified shall include, but not be limited to, the following:

- (1) Power frequency voltage withstand tests.
- (2) Voltage tests on wiring.
- (3) Resistance measurement tests on main circuit.
- (4) Mechanical operating and pressure tests.
- (5) Special type tests performed for this specification.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the following:

- (1) Shop Test Reports.
- (2) All other documentation require by applicable code sand standard.
- (3) Operating tests to check speed and mechanical performance.
- (4) Storage tank leakage tests.

(Last page of E-410)